

Udiptaman Das

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PhD student in Computer Science specializing in **Multi-Agent Systems**, **LLM Reliability**, and **Knowledge Graphs**. Expert in building end-to-end RAG pipelines and clinical knowledge extraction systems. Proven track record in reducing model hallucinations via test-time scaling and trust-aware validation layers.

TECHNICAL SKILLS

- **Languages:** Python, SQL, JavaScript/TypeScript, C++, Java, R, MATLAB, Bash.
- **AI & ML Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers.
- **LLMs & Agentic AI:** LangChain, CrewAI, AutoGen; RAG, Prompt Engineering, Fine-Tuning (LoRA/QLoRA), Evaluation (RAGAS, DeepEval).
- **Knowledge Graphs & Semantic Web:** RDFLib, Neo4j, SPARQL, OWL/RDFS, SWRL, Protégé; Ontology Alignment (SNOMED CT, LOINC, ICD, RadLex, RxNorm, UMLS, BioPortal).
- **Data, Cloud & DevOps:** PostgreSQL, MongoDB, Firebase/Firestore, Docker, Git, Vercel, REST APIs, Streamlit; GCP, AWS (S3, Lambda).

EDUCATION

University of Missouri, Kansas City <i>Pursuing PhD in Computer Science</i>	Kansas City, USA Expected May 2027
University of Missouri, Kansas City <i>MS in Data Science and Analytics</i>	Kansas City, USA July 2024
Indian Institute of Technology <i>Foundational level degree in Programming and Data Science</i>	Madras, India Dec 2021
Assam Engineering College <i>BE in Electrical Engineering</i>	Assam, India Feb 2020

RESEARCH EXPERIENCE

University of Missouri, Kansas City <i>Student researcher</i>	Aug 2023 – Present
• Multi-LLM KG Construction: Architected multi-agent LLM pipelines (Gemini, GPT-4o, Grok 3) for clinical knowledge graph construction from unstructured oncology narratives with ontology-aligned RDF/OWL encoding. • LLM Evaluation & Ontology Alignment: Evaluated fine-tuned LLMs on medical datasets with FHIR-standard ontology mapping (SNOMED CT, LOINC, ICD) following FAIR principles.	

- **Neuro-Symbolic Reasoning:** Developed image-grounded Vision Knowledge Graphs and Bayesian KG reasoning for topology-aware medical analysis and LLM safety.
- **Web Development (OHJC):** Built web applications for [Our Health Jackson County](#), [United Way of Greater Kansas City](#), and [Hispanic Development Fund](#) addressing regional health disparities.

WORK EXPERIENCE

University of Missouri, Kansas City

Graduate Teaching Assistant

Aug 2024 – Present

- Assisting with C++, Java class, lab and grading homework

Student Assistant Paraprof

Jan 2023 – Dec 2023

- Assisted with equipment and chemical inventory management, surplus equipment management, office duties, and various other jobs as assigned.

Deepneura Technologies LLP

Data Science intern

Mar 2022 – May 2022

- Object detection in gaming videos.
- Signature detection in a given image.

Dibwas Websoft Technologies LLP

Junior Data Engineer

May 2021 – Oct 2021

- Designed and improved accuracy of machine learning algorithms.
- Enhanced model performance using lift and determined KPIs with shap values.
- Conducted data visualization using Python in Jupyter Notebook (matplotlib, seaborn).

PUBLICATIONS

- Ho D.H., Das U., Ho R., & Lee Y. (2024). "Leveraging Multi-Agent Systems and Large Language Models for Diabetes Knowledge Graphs." Presented at the IEEE Big Data Conference - 4th Workshop on Knowledge Graphs and Big Data.
- Das, U., Atmakuri, K. B., Ho, D. H., Lee, C., & Lee, Y. (2026). "Clinical Knowledge Graph Construction and Evaluation with Multi-LLMs via Retrieval-Augmented Generation." arXiv:2601.01844.
- Das, U., Atmakuri, K. B., Alqarni, S., Ho, D. H., Lee, Y., Lee, C., Park, J., Kang, J., & Ha, C. S. (2026). "From Perception to Reasoning: Image-Grounded Knowledge Graphs for Topology-Aware Medical Analysis in Abdominal CT." Accepted at the 1st Workshop on Medical Reasoning (MedReasoner), CVPR 2026. <https://med-reasoner.github.io/cvpr2026/>
- KG-QUEST: Evidence-Aware and Uncertainty-Calibrated Reasoning over Multimodal Knowledge Graphs." Under review at IEEE TKDE 2026.
- "ProKG: Triplet-Level Bayesian Reasoning over Knowledge Graphs for Robust LLM Safety." Under review at ACL 2026.